UNBOUNDED STRONG-PRODUCT RATIOS FOR THE GRUNDY DOMINATION NUMBER

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Abstract. We disprove the conjecture of Brešar, Bujtás, Gologranc, Klavžar, Košmrlj, Patkós, Tuza and Vizer that the Grundy domination number of a strong product of two graphs equals the product of the Grundy domination numbers of the factors. The ratio of the two sides is unbounded, and the smallest counterexample we give is the strong product of the octahedron with a graph on eleven vertices. In the other direction, the Grundy domination number of a strong product is at most the product of the minimum ranks of the factors over any field, so the conjecture holds whenever both factors attain their minimum rank over a common field.

1. Introduction

Let G be a finite simple graph and write $N[v]$ for the closed neighbourhood of a vertex v, the set consisting of v and its neighbours. A sequence $(v_1, \dots, v_k)$ of vertices of G is *legal* if every v_i dominates a vertex that none of its predecessors dominates, that is,

$$N[v_i] \setminus \bigcup_{j<i} N[v_j] \neq \emptyset \qquad (1 \le i \le k);$$

a vertex in this difference is a *footprint* of v_i.[1] The *Grundy domination number* $\gamma_{\mathrm{gr}}(G)$ is the length of a longest legal sequence; a longest legal sequence always dominates G, since otherwise an undominated vertex could be appended. The parameter was introduced by Brešar, Gologranc, Milanič, Rall and Rizzi [7], who connected it with the domination game.

In the strong product $G \boxtimes H$ two distinct vertices (g,h) and (g',h') are adjacent when $g' \in N_G[g]$ and $h' \in N_H[h]$; equivalently $N_{G\boxtimes H}[(g,h)] = N_G[g] \times N_H[h]$. Taking a Grundy dominating sequence of G and one of H and listing their pairs in lexicographic order gives a legal sequence of $G \boxtimes H$, so

$$\gamma_{\mathrm{gr}}(G \boxtimes H) \;\ge\; \gamma_{\mathrm{gr}}(G)\,\gamma_{\mathrm{gr}}(H) \tag{1.1}$$

for all G and H [6, Proposition 21].[2] In the same paper the authors conjectured that the bound is always attained.

Conjecture 1.1 ([6, Conjecture 22])**.** *For all graphs G and H, $\gamma_{\mathrm{gr}}(G \boxtimes H) = \gamma_{\mathrm{gr}}(G)\,\gamma_{\mathrm{gr}}(H)$.*

The conjecture holds when G is a caterpillar [6] and, more generally, when G is a forest, by the work of Bell, Driscoll, Krop and Wolff [2]. A proof for all graphs was announced by Herrman and Smith [8] and later withdrawn.[3] We show that the conjecture is false and that no constant multiple of its proposed value is an upper bound.

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[1] A vertex may have several footprints; any one of them witnesses that the step is legal. The vertices of a legal sequence are distinct, since a repeated vertex dominates nothing new.

[2] If d_i footprints u_i in G and d'_j footprints u'_j in H, then (d_i, d'_j) footprints (u_i, u'_j) in the lexicographic order of the pairs, because an earlier pair (d_a, d'_b) has $a < i$, so $u_i \notin N[d_a]$, or $a = i$ and $b < j$, so $u'_j \notin N[d'_b]$.

[3] The second version of arXiv:2212.04565 (12 January 2023) carries the comment "There is an error. Some cases were not considered in the proof of Theorem 1."

Theorem 1.2. *For every $C > 0$ there are nonempty finite graphs G, H such that*

$$\gamma_{\mathrm{gr}}(G \boxtimes H) > C\, \gamma_{\mathrm{gr}}(G)\gamma_{\mathrm{gr}}(H).$$

Section 3 proves this with two graphs on the five-element subsets of $\{1, \dots, m\}$. Each factor is bounded through an incidence matrix over a finite field whose nonzero entries are exactly the closed neighbourhoods, and the diagonal of the product is an independent set. The ratio grows like m^2.

The smallest counterexamples we know have the octahedron $K_{2,2,2}$ as first factor, the complete tripartite graph with three parts of size two. For $N \geq 1$ and $r \in \{1, 2\}$ let H_N^r be the graph on $\{0, 1, \dots, N-1\}$ in which u and v are adjacent when $|u - v| \equiv r \pmod 3$. We compute the Grundy domination numbers of both families for every order.

Theorem 1.3. *For every $N \geq 4$ and every $N \geq 6$ respectively,*

$$\gamma_{\mathrm{gr}}(H_N^1) = \left\lfloor \frac{N}{3} \right\rfloor + 2, \qquad \gamma_{\mathrm{gr}}(H_N^2) = \left\lfloor \frac{N+1}{3} \right\rfloor + 3.$$

Moreover $\gamma_{\mathrm{gr}}(K_{2,2,2} \boxtimes H_N^r) \geq N$ for every $N \geq 1$ and $r \in \{1, 2\}$. Thus Conjecture 1.1 fails for $K_{2,2,2} \boxtimes H_N^1$ when $N = 11$, 13, 14 or $N \geq 15$, and for $K_{2,2,2} \boxtimes H_N^2$ when $N = 19$ or $N \geq 21$.

The smallest of these counterexamples is $K_{2,2,2} \boxtimes H_{11}^1$, with 66 vertices: the factors have Grundy domination numbers 2 and 5, while the product has a legal sequence of length 11. The proof is in Sections 4 to 6. After the first vertex of a legal sequence is chosen in a suitable circulant graph containing H_N^r, the undominated vertices form a half graph, and a count at the first selected edge, or an injection at the first selected vertex of the third residue class, gives the upper bound. Explicit sequences attain it. By Proposition 25 of [6], $\gamma_{\mathrm{gr}}(G \boxtimes H) \leq |V(G)|\, \gamma_{\mathrm{gr}}(H)$, so with the octahedron as first factor the ratio is at most 3; Theorem 1.2 must vary both factors.

Both theorems rest on the same comparison, between legal sequences and matrices. For a field $\mathbb{F}$ and a graph G let $\mathrm{mr}_{\mathbb{F}}(G)$ be the least rank of a square matrix M over $\mathbb{F}$, with rows and columns indexed by $V(G)$, such that $M_{uv} \neq 0$ exactly when $v \in N_G[u]$. Lin showed, through zero forcing, that $\gamma_{\mathrm{gr}}(G)$ is at most the least rank of a real symmetric matrix of this shape [9, Theorem 3.1]. The bound passes to strong products.

Theorem 1.4. *For every field $\mathbb{F}$ and all graphs G and H,*

$$\gamma_{\mathrm{gr}}(G \boxtimes H) \leq \mathrm{mr}_{\mathbb{F}}(G)\ \mathrm{mr}_{\mathbb{F}}(H).$$

In particular, if $\gamma_{\mathrm{gr}}(G) = \mathrm{mr}_{\mathbb{F}}(G)$ and $\gamma_{\mathrm{gr}}(H) = \mathrm{mr}_{\mathbb{F}}(H)$ for one field $\mathbb{F}$, then $\gamma_{\mathrm{gr}}(G \boxtimes H) = \gamma_{\mathrm{gr}}(G)\gamma_{\mathrm{gr}}(H)$.

So a counterexample to Conjecture 1.1 needs, for every field, a factor whose Grundy domination number is smaller than its minimum rank over that field. In Theorem 1.2 the two factors are bounded over different fields, $\mathbb{F}_2$ and $\mathbb{F}_3$, and this is forced: over a common field the product of the two bounds would also bound the product graph.

The arguments are elementary: rank bounds for matrices with prescribed support, and counting in a half graph. Counts of this kind, attached to a finite structure or a group action, also appear in the author's earlier work [3, 4, 5].

2. Rank bounds

Lemma 2.1. *Let G be a finite graph and let M be a matrix over a field $\mathbb{F}$, with rows and columns indexed by $V(G)$. Suppose that*

$$M_{uv} \neq 0 \quad \Longleftrightarrow \quad v \in N_G[u].$$

Then $\gamma_{\mathrm{gr}}(G) \leq \mathrm{rank}_{\mathbb{F}} M$. Hence $\gamma_{\mathrm{gr}}(G) \leq \mathrm{mr}_{\mathbb{F}}(G)$.

Proof. Let $(v_1, \dots, v_k)$ be legal and choose a footprint w_i for each v_i. The footprints are distinct, since w_j is not dominated by $v_1, \dots, v_{j-1}$ while $w_i \in N[v_i]$ for $i < j$. If $i < j$, then $w_j \notin N_G[v_i]$, so $M_{v_i w_j} = 0$, and each $M_{v_i w_i}$ is nonzero. The submatrix with rows $v_1, \dots, v_k$ and columns $w_1, \dots, w_k$, in this order, is lower triangular with nonzero diagonal, so its rank is k, and $k \le \operatorname{rank} M$. □

For real symmetric matrices this is the inequality of Lin cited above; the direct argument needs neither symmetry nor a particular field. The condition on M cannot be weakened to vanishing on non-edges: the footprint w_i need not equal v_i, so the entries M_{uv} with v a neighbour of u must be nonzero as well.[4]

Proof of Theorem 1.4. Let M and M' be matrices over $\mathbb{F}$ of ranks $\operatorname{mr}_{\mathbb{F}}(G)$ and $\operatorname{mr}_{\mathbb{F}}(H)$ with the supports of Lemma 2.1. The Kronecker product $M \otimes M'$ has rows and columns indexed by $V(G) \times V(H)$ and entries $M_{gg'} M'_{hh'}$. A product of two elements of a field is nonzero exactly when both are, so this entry is nonzero exactly when $g' \in N_G[g]$ and $h' \in N_H[h]$, that is, when $(g', h') \in N_{G \boxtimes H}[(g, h)]$. Lemma 2.1 and $\operatorname{rank}(M \otimes M') = \operatorname{rank} M \cdot \operatorname{rank} M'$ give the bound. If both factors attain their minimum ranks, it meets the lower bound (1.1). □

Example 2.2. Complete graphs, edgeless graphs and complete bipartite graphs attain their minimum ranks over $\mathbb{C}$, so Conjecture 1.1 holds for every strong product of two such graphs. For K_n the all-ones matrix has rank $1 = \gamma_{\mathrm{gr}}(K_n)$, and for the edgeless graph the identity matrix has rank $|V| = \gamma_{\mathrm{gr}}$. For $K_{a,b}$ with $a \ge b \ge 1$, the larger side is independent, so listing it gives a legal sequence and $\gamma_{\mathrm{gr}}(K_{a,b}) \ge a$. With $\zeta = e^{2\pi i/a}$, the block matrix

$$M = \begin{pmatrix} aI_a & Z \\ Z^* & I_b \end{pmatrix}, \qquad Z_{kj} = \zeta^{kj} \quad (0 \le k < a,\ 0 \le j < b),$$

has the support of Lemma 2.1. Since $Z^*Z = aI_b$, its last b rows are $\frac{1}{a}Z^*$ times its first a rows, so $\operatorname{rank} M = a$ and $\gamma_{\mathrm{gr}}(K_{a,b}) = a = \operatorname{mr}_{\mathbb{C}}(K_{a,b})$.

3. Unbounded product ratios

To obtain unbounded ratios we vary both factors. The factor bounds come from incidence matrices over different fields, in the manner of the intersection-congruence arguments used by Alon [1, Section 3] for the Shannon capacity; the application here needs the nonzero entries to be exactly the closed neighbourhoods, so that Lemma 2.1 applies. No result about the Shannon capacity is used.

Theorem 3.1. *For $m \ge 5$, let A_m and B_m have vertex set $\binom{[m]}{5}$, the five-element subsets of $[m] = \{1, \dots, m\}$. Distinct sets S, T are adjacent in A_m when $|S \cap T|$ is odd, and in B_m when $|S \cap T| \equiv 2 \pmod 3$. Then*

$$\gamma_{\mathrm{gr}}(A_m) \le m, \qquad \gamma_{\mathrm{gr}}(B_m) \le \binom{m}{2}, \qquad \gamma_{\mathrm{gr}}(A_m \boxtimes B_m) \ge \binom{m}{5}.$$

In particular,

$$\frac{\gamma_{\mathrm{gr}}(A_m \boxtimes B_m)}{\gamma_{\mathrm{gr}}(A_m)\gamma_{\mathrm{gr}}(B_m)} \ge \frac{(m-2)(m-3)(m-4)}{60m}, \tag{3.1}$$

so the ratios are unbounded.

Proof. Let U be the incidence matrix between five-element subsets of $[m]$ and elements of $[m]$, with $U_{Sx} = 1$ when $x \in S$. Over $\mathbb{F}_2$,

$$(UU^{\mathsf{T}})_{ST} = |S \cap T| \pmod 2.$$

[4]For an independent set the diagonal alone would do; this is the difference between bounding the independence number, as in the Shannon capacity arguments of Alon [1], and bounding the Grundy domination number.

Since $|S| = 5$ is odd, the diagonal entries are nonzero, and an off-diagonal entry is nonzero exactly when $|S \cap T|$ is odd, that is, when S and T are adjacent in A_m. So the nonzero entries of UU^{T} are exactly the closed neighbourhoods of A_m. Its rank is at most the number m of columns of U, and Lemma 2.1 gives the first bound.

Let V be the incidence matrix between five-element subsets and two-element subsets of $[m]$, with $V_{SP} = 1$ when $P \subseteq S$. Over $\mathbb{F}_3$,

$$(VV^{\mathsf{T}})_{ST} = \binom{|S \cap T|}{2} \pmod 3,$$

the number of two-element subsets of $S \cap T$. For intersection sizes $t = 0, 1, 2, 3, 4, 5$ the values $\binom{t}{2} = 0, 0, 1, 3, 6, 10$ reduce to $0, 0, 1, 0, 0, 1$ modulo 3. So the entry is nonzero exactly when $S = T$ or $|S \cap T| = 2$, which are precisely the closed neighbourhoods of B_m. The rank is at most $\binom{m}{2}$, proving the second bound.

The product vertices

$$\mathcal{D}_m = \{(S, S) : S \in \tbinom{[m]}{5}\}$$

form an independent set of $A_m \boxtimes B_m$: for $S \neq T$, adjacency of (S, S) and (T, T) would need S and T adjacent in both factors, which would require $|S \cap T|$ to be 1 or 3 and also 2. Listing the vertices of $\mathcal{D}_m$ in any order gives a legal sequence, each vertex footprinting itself, because no earlier vertex of the list dominates it. This proves the lower bound $\binom{m}{5}$ for the product.

The three bounds give (3.1), since

$$\begin{aligned}
&\binom{m}{5} \Big/ \Big(m\binom{m}{2}\Big) \\
&= \frac{m(m-1)(m-2)(m-3)(m-4)}{120} \cdot \frac{2}{m^2(m-1)} \\
&= \frac{(m-2)(m-3)(m-4)}{60m},
\end{aligned}$$

and the right side tends to infinity with m. □

Proof of Theorem 1.2. Take $G = A_m$ and $H = B_m$ with m so large that $(m-2)(m-3)(m-4) > 60Cm$. □

For example, $m = 13$ gives $\gamma_{\mathrm{gr}}(A_{13} \boxtimes B_{13}) \geq 1287$, whereas $\gamma_{\mathrm{gr}}(A_{13})\gamma_{\mathrm{gr}}(B_{13}) \leq 13 \cdot 78 = 1014$. The two factor bounds use different fields; no matrix product over a common field is involved, and nothing is claimed about the exact values of $\gamma_{\mathrm{gr}}(A_m)$ and $\gamma_{\mathrm{gr}}(B_m)$.

4. The octahedron and the graphs H_N^r

We write the vertices of the octahedron as $g_0, g_1, g_2, x_0, x_1, x_2$, where x_t is the antipode of g_t; the three antipodal pairs are the only non-adjacent pairs (Figure 1). Thus $N[v]$ is everything except the antipode of v.

Lemma 4.1. $\gamma_{\mathrm{gr}}(K_{2,2,2}) = 2$.

Proof. The sequence (g_0, x_0) is legal: x_0 footprints itself. Conversely, let (u, v) be legal. If v is the antipode of u, then $N[u] \cup N[v]$ is the whole vertex set; otherwise $N[u]$ misses only the antipode of u, which lies in $N[v]$. Either way nothing is left for a third vertex to footprint. □

In H_N^r we write $a_i = 3i$, $b_i = 3i + 1$ and $c_i = 3i + 2$. Two vertices of the same residue class modulo 3 are never adjacent. For vertices of different classes, a difference $3k + e$ with $k < 0$ and $e \in \{1, 2\}$ has absolute value $3|k| - e \equiv 3 - e \pmod 3$, and this gives

$$\text{in } H_N^1: \quad a_i \sim b_j \iff j \geq i, \qquad b_i \sim c_j \iff j \geq i, \qquad c_i \sim a_j \iff j > i, \tag{4.1}$$

$$\text{in } H_N^2: \quad a_i \sim c_j \iff j \geq i, \qquad b_i \sim a_j \iff j > i, \qquad c_i \sim b_j \iff j > i, \tag{4.2}$$

from $b_j - a_i = c_j - b_i = 3(j - i) + 1$ and $a_j - c_i = 3(j - i - 1) + 1$ in the first line, and $c_j - a_i = 3(j - i) + 2$ and $a_j - b_i = b_j - c_i = 3(j - i - 1) + 2$ in the second.

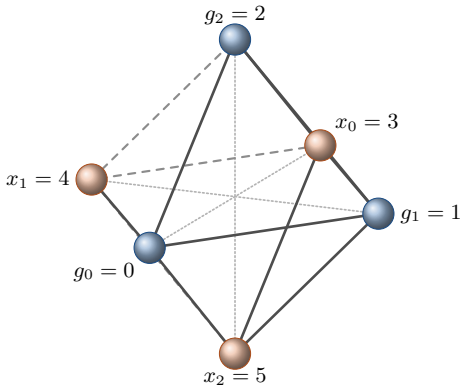


FIGURE 1. The octahedron $K_{2,2,2}$. Each vertex is adjacent to all others except its antipode; the dotted segments join the antipodal pairs $\{g_t, x_t\}$.

5. A long legal sequence in the product

The construction uses one property of H_N^r. Colour v by $t(v) = -v \bmod 3$ in H_N^1 and by $t(v) = v \bmod 3$ in H_N^2. If uv is an edge with $u < v$, then $v - u \equiv r \pmod 3$, and in both cases $t(u) \equiv t(v) + 1 \pmod 3$. We state the construction for any graph with this property.

Proposition 5.1. *Let H be a graph with vertices $v_0, \dots, v_{N-1}$ and a map $t\colon V(H) \to \mathbb{Z}_3$ such that $t(v_i) \equiv t(v_j) + 1 \pmod 3$ for every edge v_iv_j with $i < j$. Then the sequence*

$$s_j = (g_{t(v_j)},\, v_j), \qquad j = 0, 1, \dots, N-1,$$

is legal in $K_{2,2,2} \boxtimes H$, and $\gamma_{\mathrm{gr}}(K_{2,2,2} \boxtimes H) \geq N$. In particular $\gamma_{\mathrm{gr}}(K_{2,2,2} \boxtimes H_N^r) \geq N$ for every $N \geq 1$ and $r \in \{1, 2\}$.

Proof. We show that s_j footprints $f_j = (x_{t(v_j)+1}, v_j)$, the indices of g and x being taken modulo 3. First, $f_j \in N[s_j]$: the vertex $x_{t(v_j)+1}$ is not the antipode $x_{t(v_j)}$ of $g_{t(v_j)}$, so it lies in $N[g_{t(v_j)}]$, and $v_j \in N[v_j]$. Now let $i < j$. If $f_j \in N[s_i]$, then $x_{t(v_j)+1} \in N[g_{t(v_i)}]$, which means $t(v_i) \neq t(v_j) + 1$, and $v_j \in N[v_i]$, which for $i \neq j$ means that v_iv_j is an edge, hence $t(v_i) = t(v_j) + 1$. The two conclusions contradict each other, so $f_j \notin N[s_i]$. □

Figure 2 shows the sequence for H_{21}^2. Every column of the product contains exactly one vertex of the sequence, while a sequence of the form used for (1.1) uses only $\gamma_{\mathrm{gr}}(H_{21}^2) = 10$ columns. The rotation of the three colours along the sequence lets each new vertex reach a footprint in a layer x_t that none of the earlier vertices covers in that column.

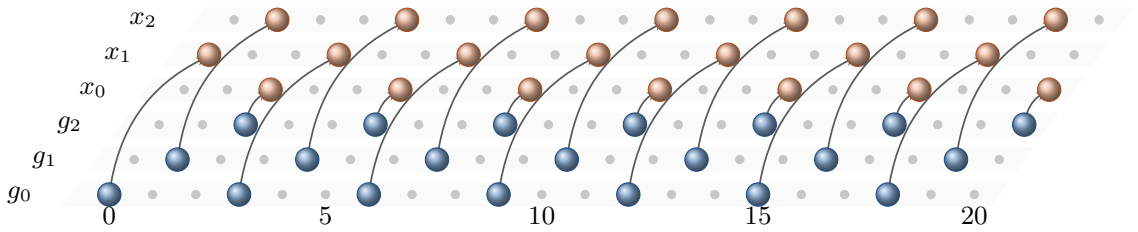


FIGURE 2. $K_{2,2,2} \boxtimes H_{21}^2$ as six stacked copies of H_{21}^2, one for each vertex of the octahedron. The 21 blue vertices $(g_{j \bmod 3}, j)$ form a legal sequence; the arrow from each of them ends at its footprint $(x_{(j+1) \bmod 3}, j)$.

6. The Grundy domination numbers of H_N^1 and H_N^2

A legal sequence in an induced subgraph remains legal in the larger graph, with the same footprints, since closed neighbourhoods in the subgraph are the intersections of those in the larger graph with the subgraph. We use this observation for both bounds. The upper bounds come from one lemma about half graphs.

Lemma 6.1. *Let $q \geq 1$, $\varepsilon \in \{0,1\}$, and let z be a vertex of a graph Γ whose non-neighbours are $2q$ vertices X_i, Y_j $(0 \leq i, j < q)$. Suppose that there are no edges among the X_i and none among the Y_j, that*

$$X_i \sim Y_j \quad \Longleftrightarrow \quad i \geq j, \tag{6.1}$$

and that every neighbour w of z has an index $k \in \{0, \dots, q\}$ with

$$N[w] \cap \{X_i, Y_j\} = \{X_i : i < k\} \cup \{Y_j : k + \varepsilon \leq j < q\}. \tag{6.2}$$

Then every legal sequence of Γ that begins with z has length at most $q + 2 + \varepsilon$.

Proof. The vertex z dominates every vertex except the X_i and Y_j, and the $2q$ vertices X_i, Y_j induce a half graph with the perfect matching X_iY_i (Figure 3). Follow the sequence from its second vertex on until either a neighbour of z is selected or the selected X- and Y-vertices contain an edge for the first time. Before either event, let I and J be the sets of selected indices on the two sides, and put $r = |I|$, $s = |J|$. By (6.1), the absence of edges means

$$i < j \qquad (i \in I,\ j \in J). \tag{6.3}$$

Each legal step dominates at least one vertex that was undominated, so the number of steps after a given moment is at most the number of vertices undominated at that moment. We treat the possible events in turn.

A new X_i creates an edge. Then $J \neq \emptyset$ and $b = \min J \leq i$. All earlier X-indices are less than b by (6.3), so i is the largest selected X-index. After this step, an X-vertex is dominated if it was selected or if it is adjacent to a selected Y_j, that is, if its index is at least b; so the undominated X-indices are $\{0, \dots, b-1\} \setminus I$. A Y-vertex is dominated if it was selected or is adjacent to a selected X-vertex, and the selected X-vertex with the largest index is X_i; so the undominated Y-indices are those greater than i that are not in J. Writing $s_0 = |J \cap \{b, \dots, i\}|$ for the number of selected Y-indices at most i, the number of undominated vertices is

$$(b - r) + (q - 1 - i) - (s - s_0).$$

The number of vertices selected after z, which is $r + s + 1$, plus the number still undominated, is

$$r + s + 1 + b - r + q - 1 - i - s + s_0 = q + b - i + s_0 \leq q + 1,$$

because $s_0 \leq i - b + 1$. The full sequence, including z, has length at most $q + 2$.

A new Y_j creates an edge. The map $X_i \mapsto Y_{q-1-i}$, $Y_j \mapsto X_{q-1-j}$ exchanges the two sides and preserves (6.1), since $q - 1 - j \geq q - 1 - i$ exactly when $i \geq j$. Up to this step only X- and Y-vertices were selected after z, so the count of the previous case applies, and again the length is at most $q + 2$.

A neighbour w of z is selected first. Let k be its index in (6.2). After this step every vertex still undominated belongs to

$$W_k = \{X_i : k \leq i < q\} \cup \{Y_j : 0 \leq j < \min(k + \varepsilon, q)\}, \qquad |W_k| \leq q + \varepsilon.$$

At least $r + s$ vertices of W_k were dominated already before w was selected. To see this, map each earlier selection by

$$X_i \longmapsto \begin{cases} X_i, & i \geq k, \\ Y_i, & i < k, \end{cases} \qquad Y_j \longmapsto \begin{cases} Y_j, & j < k + \varepsilon, \\ X_j, & j \geq k + \varepsilon. \end{cases}$$

Each image lies in W_k and is either the selected vertex itself or its partner in the matching, which the selected vertex dominates by (6.1). Images of distinct selections on the same side

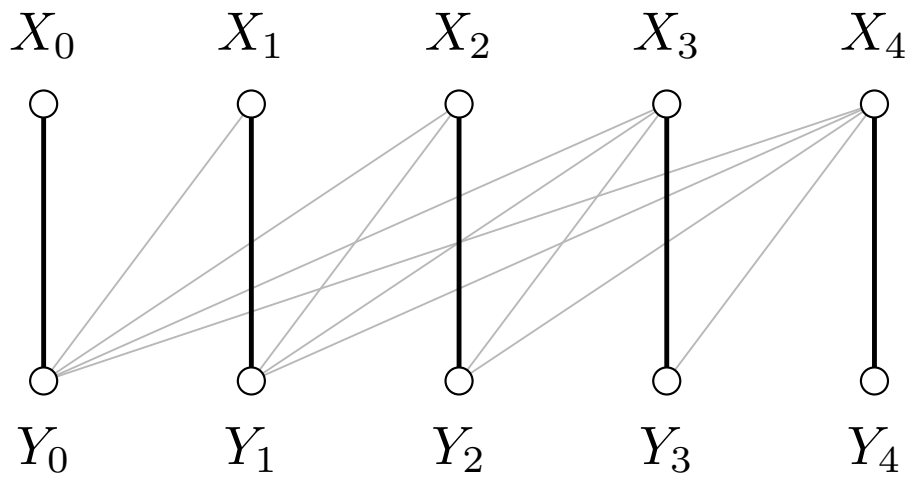


FIGURE 3. The non-neighbours of the first vertex $z = 0$ in H^1_{3q+2} and in H^2_{3q+1}, drawn for $q = 5$. The matching edges X_iY_i are emphasised. Before the first selected edge, every selected X-index is smaller than every selected Y-index.

are distinct. An image shared by a selection X_i and a selection Y_j would force $i = j$, with $i \in I$ and $j \in J$, contrary to (6.3). So the map is injective, at most $q + \varepsilon - r - s$ vertices remain undominated after w, and the sequence has length at most

$$1 + (r + s) + 1 + (q + \varepsilon - r - s) = q + 2 + \varepsilon.$$

Neither event occurs. Then the vertices selected after z form an independent set of the half graph, which has the perfect matching X_iY_i, so there are at most q of them and the sequence has length at most $q + 1$. □

Lemma 6.2. *For every integer $q \geq 1$, $\gamma_{\mathrm{gr}}(H^1_{3q+2}) \leq q + 2$ and $\gamma_{\mathrm{gr}}(H^2_{3q+1}) \leq q + 3$.*

Proof. Let $r \in \{1, 2\}$ and $n = 3q + 3 - r$, that is, $n = 3q + 2$ for $r = 1$ and $n = 3q + 1$ for $r = 2$. In both cases $n \equiv 2r \pmod 3$, so for $1 \leq d < n$ the conditions $d \equiv r$ and $n - d \equiv r \pmod 3$ are equivalent. Hence u and v are adjacent in H^r_n exactly when $(u - v) \bmod n$ is congruent to r modulo 3, and translation modulo n preserves adjacency. A legal sequence stays legal under a graph automorphism, so we may translate a nonempty legal sequence so that its first vertex is $z = 0$. In both graphs put

$$X_i = a_{i+1} = 3(i + 1), \qquad 0 \leq i < q.$$

In H^1_{3q+2} the neighbours of 0 are the $b_k = 3k + 1$, $0 \leq k \leq q$, and its non-neighbours are the X_i and the vertices $Y_j = c_j = 3j + 2$, $0 \leq j < q$. By (4.1), $X_i = a_{i+1} \sim c_j$ exactly when $i + 1 > j$, which is (6.1); and b_k is adjacent to a_{i+1} exactly when $i + 1 \leq k$ and to c_j exactly when $j \geq k$, which is (6.2) with $\varepsilon = 0$. Lemma 6.1 gives $q + 2$.

In H^2_{3q+1} the neighbours of 0 are the $c_k = 3k + 2$, $0 \leq k < q$, and its non-neighbours are the X_i and the vertices $Y_j = b_j = 3j + 1$, $0 \leq j < q$. By (4.2), $a_{i+1} \sim b_j$ exactly when $i + 1 > j$, which is (6.1); and c_k is adjacent to a_{i+1} exactly when $i + 1 \leq k$ and to b_j exactly when $j > k$, which is (6.2) with $\varepsilon = 1$. Lemma 6.1 gives $q + 3$. □

Lemma 6.3. *If $N \geq 4$ and $q = \lfloor N/3 \rfloor$, then $\gamma_{\mathrm{gr}}(H^1_N) \geq q + 2$.*

Proof. For $q = 1$, the sequence $(0, 1, 2)$ in H^1_4, a path, has the footprints $0, 2, 3$. For $q = 2$, the sequence $(0, 5, 1, 2)$ in H^1_6 has the footprints $0, 5, 2, 3$. For $q \geq 3$ we work in H^1_{3q}, an induced

subgraph of H_N^1, whose vertices are a_i, b_i, c_i for $0 \le i < q$, and take

$$a_0,\ a_1,\ c_0,\ c_1,\ c_2, \dots, c_{q-2},\ b_0,$$

of length $q+2$. By (4.1), a_0 dominates itself and every b_j, and a_1 dominates itself and c_0; the vertex c_0 then footprints a_2, which neither a_0 nor a_1 dominates. For $1 \le m \le q-2$ the vertex c_m is adjacent to no a_i with $i \le 1$ and to no other c, so it footprints itself. Finally b_0 is adjacent to every c_j, and c_{q-1} has not been dominated, so it is the footprint of b_0. □

Lemma 6.4. *If $N \ge 6$ and $q = \lfloor (N+1)/3 \rfloor$, then $\gamma_{\mathrm{gr}}(H_N^2) \ge q+3$.*

Proof. For $q = 2$, the sequence $(1, 3, 5, 0, 2)$ in H_6^2 has the footprints $1, 5, 0, 2, 4$ and remains legal in H_7^2. For $q = 3$, the sequence $(0, 7, 2, 4, 5, 1)$ in H_8^2 has the footprints $0, 7, 4, 6, 3, 1$ and remains legal in H_9^2 and H_{10}^2. Each footprint can be read off from (4.2).

For $q \ge 4$ it suffices to work in H_{3q-1}^2, which is an induced subgraph of H_N^2 because $3q-1 \le N$. Its vertices are a_i, b_i for $0 \le i \le q-1$ and c_i for $0 \le i \le q-2$. Take the sequence

$$a_0, a_1, \dots, a_{q-3},\ b_{q-1},\ c_{q-3},\ c_0,\ a_{q-2},\ b_0. \tag{6.4}$$

The first $q-2$ vertices footprint themselves, since vertices of one class are never adjacent. The last five have the successive footprints

$$b_{q-1}, \quad b_{q-2}, \quad b_{q-3}, \quad a_{q-2}, \quad a_{q-1},$$

by (4.2): b_{q-1} is adjacent to no a_i with $i \le q-3$, so it is undominated and footprints itself; b_{q-2} is adjacent to c_{q-3} and not to any earlier vertex of the sequence; b_{q-3} is adjacent to c_0, because $q-3 > 0$, and not to $a_0, \dots, a_{q-3}$, b_{q-1} or c_{q-3}; neither a_{q-2} nor a_{q-1} is adjacent to b_{q-1}, c_{q-3} or c_0, so a_{q-2} footprints itself and b_0 footprints a_{q-1}. The sequence has length $(q-2)+5 = q+3$. □

Proof of Theorem 1.3. For H_N^1 with $N \ge 4$ put $q = \lfloor N/3 \rfloor \ge 1$. Then $N \le 3q+2$, so H_N^1 is an induced subgraph of H_{3q+2}^1, and Lemmas 6.2 and 6.3 give $\gamma_{\mathrm{gr}}(H_N^1) = q+2$. For H_N^2 with $N \ge 6$ put $q = \lfloor (N+1)/3 \rfloor \ge 2$. Then $N \le 3q+1$, and Lemmas 6.2 and 6.4 give $\gamma_{\mathrm{gr}}(H_N^2) = q+3$. The product bound is Proposition 5.1.

The conjecture fails for $K_{2,2,2} \boxtimes H_N^r$ whenever $N > 2\gamma_{\mathrm{gr}}(H_N^r)$, by Lemma 4.1. For $r = 1$ write $N = 3q+e$ with $e \in \{0, 1, 2\}$; then $N - 2(q+2) = q+e-4$ is positive exactly when $q \ge 5-e$, that is, for $N = 11$, 13, 14 and every $N \ge 15$. For $r = 2$ write $N = 3q-1$, $3q$ or $3q+1$; the differences $N - 2(q+3)$ are $q-7$, $q-6$ and $q-5$, positive for $N = 19$ and every $N \ge 21$. □

Example 6.5. For $N = 11$ the sequence of Lemma 6.3 with $q = 3$ is $(0, 3, 2, 5, 1)$, of length $5 = \gamma_{\mathrm{gr}}(H_{11}^1)$, while $K_{2,2,2} \boxtimes H_{11}^1$, with 66 vertices, has the legal sequence $s_j = (g_{(-j) \bmod 3}, j)$, $0 \le j \le 10$, of length 11; the footprint of s_j is $(x_{(1-j) \bmod 3}, j)$. For $N = 19$ the sequence (6.4) with $q = 6$ is $(0, 3, 6, 9, 16, 11, 2, 12, 1)$, of length $9 = \gamma_{\mathrm{gr}}(H_{19}^2)$, and $K_{2,2,2} \boxtimes H_{19}^2$ has 114 vertices and a legal sequence of length 19. For $N = 12$ and $r = 1$, and for $N = 20$ and $r = 2$, the formula gives exactly $N = 2\gamma_{\mathrm{gr}}(H_N^r)$, and Theorem 1.3 says nothing.

Remark 6.6. The following values were found by exhaustive computer search and are not used in the proofs. For $4 \le N \le 13$, $\gamma_{\mathrm{gr}}(K_{2,2,2} \boxtimes H_N^1)$ equals $6, 6, 8, 8, 8, 10, 10, 11, 12, 13$, which is $2\gamma_{\mathrm{gr}}(H_N^1)$ for $N \le 10$ and $N = 12$; and $\gamma_{\mathrm{gr}}(K_{2,2,2} \boxtimes H_N^2) = 2\gamma_{\mathrm{gr}}(H_N^2)$ for $6 \le N \le 12$. So the length N of Proposition 5.1 is optimal for $K_{2,2,2} \boxtimes H_{11}^1$ and $K_{2,2,2} \boxtimes H_{13}^1$.

Remark 6.7. Everything above is proved in full. Theorem 1.2 rests on Lemma 2.1 and the two incidence matrices; Theorem 1.3 on the half-graph count of Lemma 6.1, the explicit sequences of Lemmas 6.3 and 6.4, and Proposition 5.1. The results quoted from elsewhere are the inequality (1.1) and the bound $\gamma_{\mathrm{gr}}(G \boxtimes H) \le |V(G)|\gamma_{\mathrm{gr}}(H)$, both from [6] with proofs of a few lines, and Lin's inequality, which Lemma 2.1 reproves; none of them is needed for the counterexamples. Nothing is claimed about the exact values of $\gamma_{\mathrm{gr}}(A_m)$, $\gamma_{\mathrm{gr}}(B_m)$ or $\gamma_{\mathrm{gr}}(A_m \boxtimes B_m)$ beyond the three bounds of Theorem 3.1, nor about the smallest possible counterexample.

7. Questions

Question 7.1. Which graphs G satisfy Conjecture 1.1 for every H? All forests do [2], and by Theorem 1.4 so does every G with $\gamma_{\mathrm{gr}}(G) = \mathrm{mr}_{\mathbb{F}}(G)$ for every field $\mathbb{F}$, such as complete and edgeless graphs; the octahedron does not.

Question 7.2. Is there a counterexample to Conjecture 1.1 with fewer than 66 vertices?

Declarations

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Conflict of interest. None.

Data availability. The source of this paper and programs that recompute $\gamma_{\mathrm{gr}}(H_N^r)$ and $\gamma_{\mathrm{gr}}(K_{2,2,2} \boxtimes H_N^r)$ for small N, the values of Remark 6.6 and the sequences of Section 6 are at `https://github.com/creelie/Paper_1`, release `v2.1.0`, archived on Zenodo under the DOI 10.5281/zenodo.23791279 (all versions: 10.5281/zenodo.23607651). The paper does not depend on them. The programs were written and run by the author with the assistance of Claude Code.

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